

L'ORÉAL – Batch Decoding & Automatic DFAB/DLUO Calculation

System: Speed WMS

Speed Version: A.8.6.F (28.06b1)

Radio Version: 28.06b

Module STD175: 28.06

1. Document Control

1.1 Version History

Version Description		Date	Author
1.0	Initial version	03/04/2025	AKD
1.1	Added batch decoding for cycle counting	07/04/2025	AKD
1.2	Email feedback update	09/04/2025	AKD
1.3	Teams exchange update	17/04/2025	AKD

2. Business Context

Within the **L'ORÉAL activity**, the **Multi-BL Reception module** is used to receive articles.

A control must be implemented to:

Decode the received batch number (LOT)

Automatically determine:

- Manufacturing Date (DFAB)
- Best Before Date (DLUO)

△ The decoding logic depends on how the batch number is entered:

- Manually entered
 - Retrieved from standard EAN128 scan
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3. Functional Design

3.1 Activation Parameter

Control is applied only if:

ACT_TOP18 = 1

If ACT_TOP18 ≠ 1:

→ No automatic decoding is performed.

4. Batch Decoding Rules

There are **two distinct decoding methods**:

4.1 Method 1 – "Before ISIS"

Condition

The batch number:

- Starts with "+"
- Has the same length as ACT_ALPHA1
- Mask defined in activity IC:

ACT_ALPHA1 = "+AJJX"

4.1.1 Mask Interpretation

The mask is dynamic.

We must retrieve:

- Position of first "A" (Year)
- Number of consecutive "A"
- Position of "JJJ" (Day of year)

⚠ Positions are NOT fixed — must be dynamically extracted from ACT_ALPHA1.

Year (A)

- Extract position of "A"
- Use transcoding table where:

TRANS_NAME = "AYR"

Example:

K → 2013

Day of Year (JJJ)

- Extract numeric day (001–365 or 366)
- Must respect leap year logic

Examples:

001 → 01/01

365 → 31/12 (non-leap year)

366 → 31/12 (leap year)

Example – Case 1

Batch:

+K0060

Article:

ART_NUM1 = 10

From TRANS table:

K → 2013

006 → 6th day of year

DFAB:

06 January 2013

DLUO:

DFAB + ART_NUM1 days

06/01/2013 + 10 = 16/01/2013

4.2 Method 2 – "After ISIS"

Condition

The batch:

- Does NOT start with "+"

- Contains at least one letter
- Has same length as ACT_ALPHA2
- Mask defined in:

ACT_ALPHA2 = XXAMXX

4.2.1 Mask Interpretation

Positions must be dynamically extracted.

Retrieve:

- Position of "A" → Year
 - Position of "M" → Month
-

Year (A)

Use transcoding table:

TRANS_NAME = "BYR"

Example:

J → 2012

Month (M)

Use transcoding table:

TRANS_NAME = "AMO"

Example:

7 → July

O → October

N → November

D → December

Default rule:

Manufacturing date = First day of extracted month

Example – Case 2

Batch:

2YJ700

Article:

ART_NUM1 = 10

From TRANS table:

J → 2012

7 → July

DFAB:

01 July 2012

DLUO:

01/07/2012 + 10 days = 11/07/2012

4.3 Invalid Batch

If the batch:

- Does not start with "+"
- Does not match ACT_ALPHA2 format
- Does not contain required letter positions

→ It cannot be decoded.

In this case:

- DFAB must be manually entered
 - DLUO must be manually entered
 - DLC remains manually entered (no relation to batch)
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5. DLUO Calculation Rule

Once DFAB is determined:

DLUO = DFAB + ART_NUM1 (Number of days between DFAB and DLUO)

6. Functional Flow – Multi-BL Reception

1. Select Multi-BL Reception
2. Select Reception / BL
3. Flash Support
4. Flash Article / EAN
5. Enter Quantity
6. Open Traceability Window
7. Batch1 field:
 - If EAN scanned → batch auto-filled
 - Else → operator manually enters batch

Control Point

Use:

PE882

At initialization of traceability window.

If ACT_TOP18 = 1:

→ Perform decoding logic.

7. Technical Implementation

7.1 PE882 – Reception

- Duplicate standard window into specific version
- Perform decoding logic inside specific window
- DLC remains manual

Check:

ACT_TOP18 = 1

Then:

Logic Summary

IF LOT starts with "+" AND matches ACT_ALPHA1 length:

- Decode using AYR
- Extract JJJ
- Calculate DFAB
- Calculate DLUO

ELSE IF LOT matches ACT_ALPHA2:

- Decode using BYR
- Extract month using AMO
- DFAB = First day of month
- Calculate DLUO

ELSE:

- No decoding
- Manual entry required

7.2 PE884 – Cycle Counting (Version 1.2)

Same decoding logic must apply for:

- Cycle Counting module

Implementation:

- Duplicate window
- Apply same logic as PE882

8. Transcoding Table – TRANS_BAS

Table structure:

Field	Type	Example
TRANS_NAME	String	AYR
TRANS_IN	String	K

Field	Type	Example
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TRANS_OUT	String	2013
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8.1 Year Transcoding (AYR & BYR)

Letters I, O, Q are excluded.

Example mapping (valid until 31/12/2025):

A → 2004

B → 2005

...

K → 2013

...

Z → 2026

8.2 Month Transcoding (AMO)

1 → January

2 → February

...

9 → September

O → October

N → November

D → December

8.3 Year Mapping From 01/01/2026

New mapping applies:

A → 2026

B → 2027

...

Z → 2048

Must be updated in TRANS_BAS table accordingly.

9. Client Validation

AGL acknowledges and validates the content of this document.

BK Systèmes commits to delivering the solution according to the approved scope.

Any additional evolution outside this scope may require additional quotation.

Signed at: Vaulx-Milieu

Date: 03/04/2025

Company Signature & Stamp
